LiDAR Reflection Recovery via Shadow Boxing

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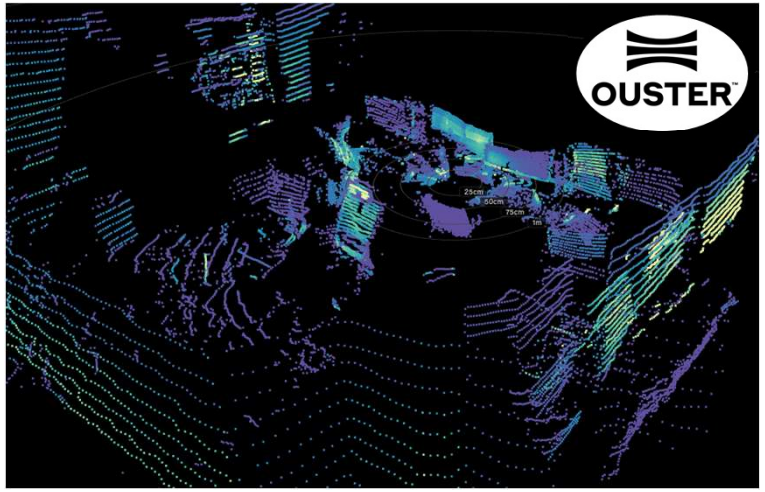
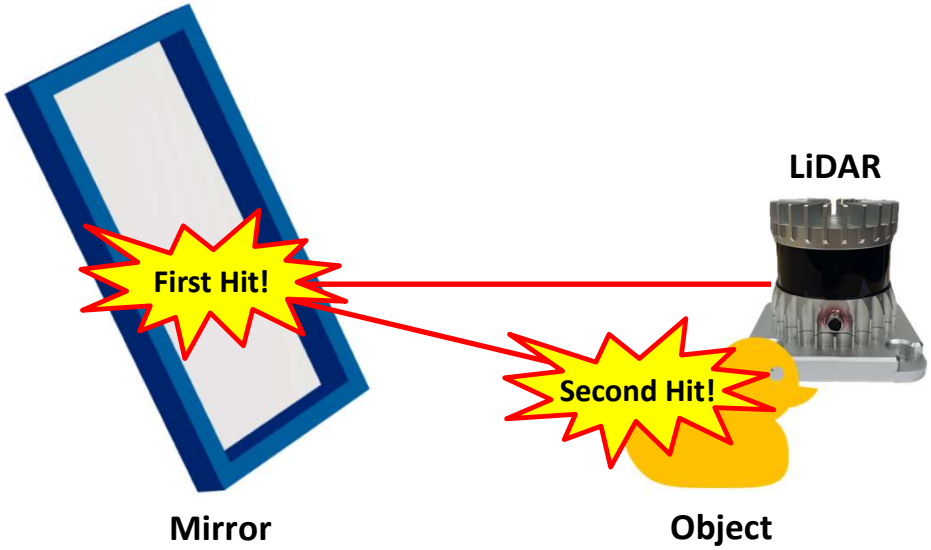
LiDAR vs Mirror

- **Light Detection And Ranging (LiDAR)**
  - A sensor that acquires spatial information using laser pulses
    - Used in robotics and autonomous driving
  - Our device is the Ouster OS1-32

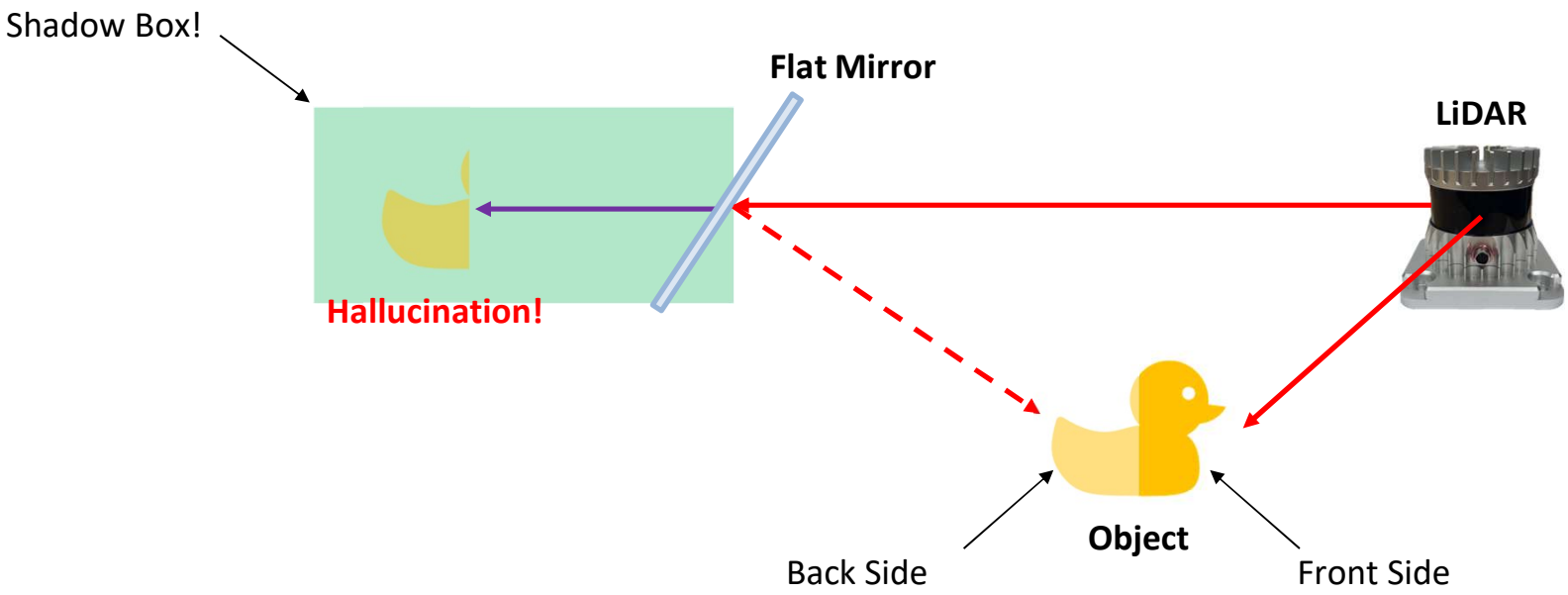


- **Mirror**
  - A challenge for LiDAR laser pulses
  - It causes reflection
    - Lasers can hit multiple objects
    - Each hit has differences in timing and intensity



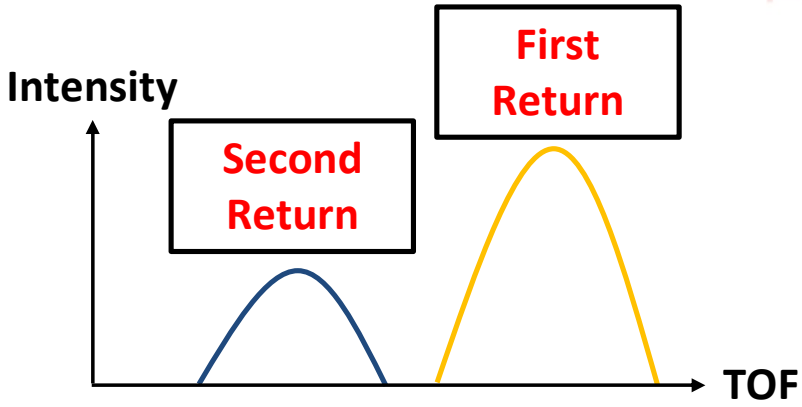
Specular Reflection

- Specular Reflection
  - Flat Mirrors cause **Specular Reflection**
  - But LiDAR sensors cannot detect direction's change
    - Perceived as passing through the mirror
- Utilization of Specular Reflection
  - Chance to capture the **sides or back of an object**
  - By concatenating, we can obtain complete 3D shape
  - Generate **Shadow Box** to detect Hallucination side

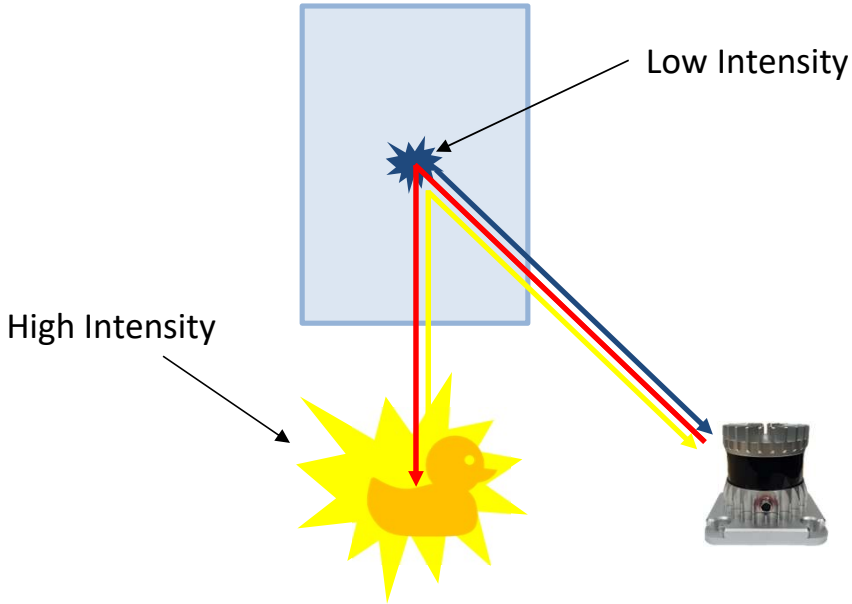


Dual Return Mode

- Dual Return Mode
  - Divide multiple returns based on a standard
    - Standard: Time Of Flight (TOF), Intensity or others
  - Ouster uses intensity as the standard



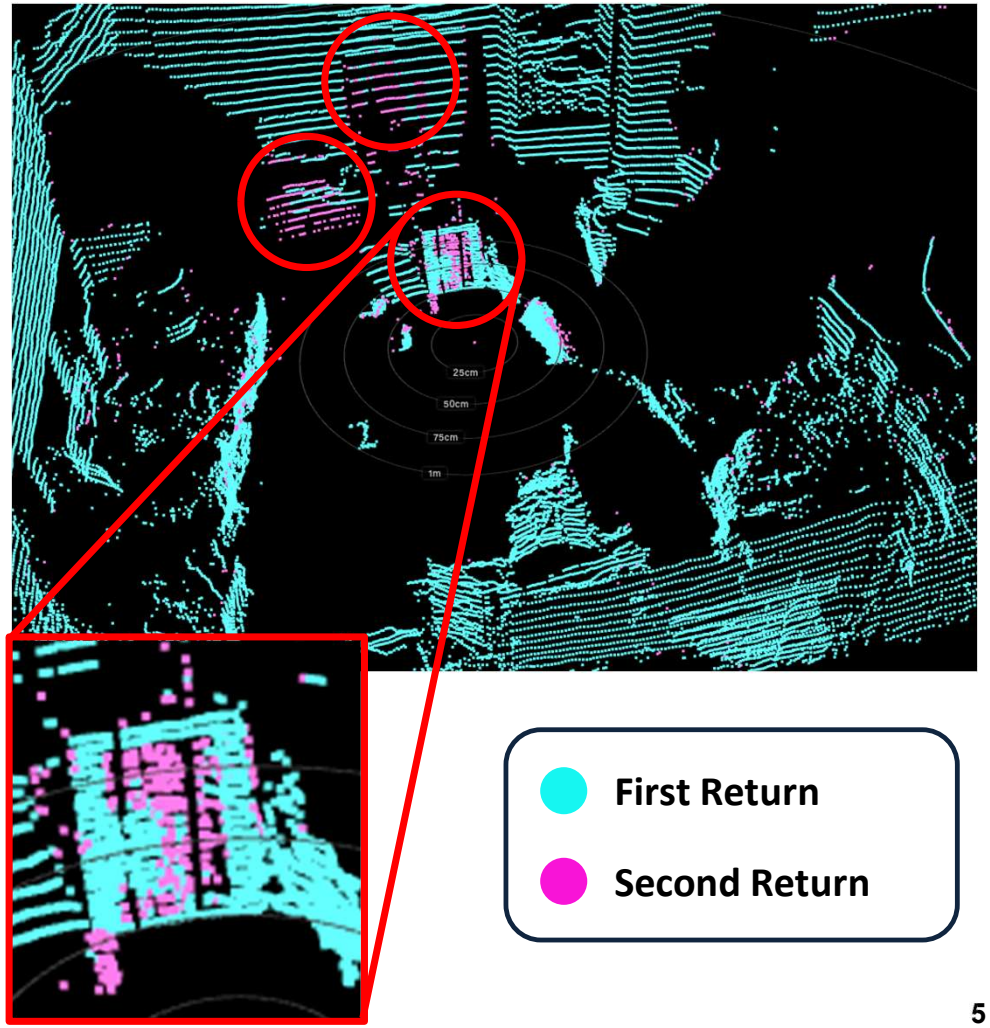
- Example: Case of Mirror
  - First, the laser hits the Mirror
    - With slight energy loss
  - Second, the laser hits the Object
    - With most energy concentrated on it



Mirror Plane Detection

- Ouster Dual Return
  - Mirror frame is a normal material **(= First Return)**
  - Mirror plane is a reflective material **(= Second Return)**
    - But quantity is only 0.4%!

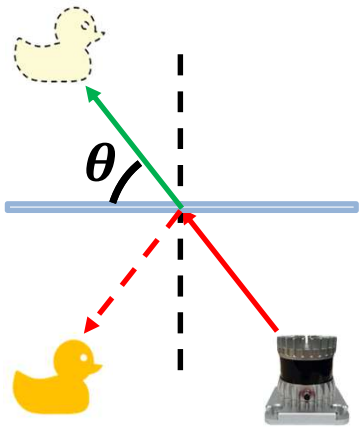
- Mirror Plane Detection Steps
  - DBSCAN Clustering
    - Utilize the 0.4% Second Returns to detect mirror's location
  - PCA
    - First planarity filter
    - Use both returns to remove noises
  - RANSAC
    - Second planarity filter
    - Robust and accurate algorithm to finalize flat mirros



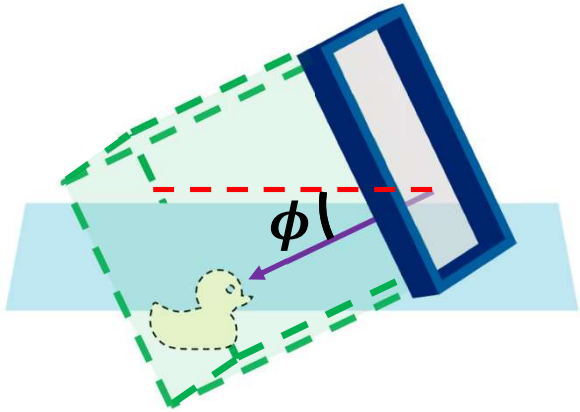
Shadow Boxing Method

- Shadow Box Direction
  - Horizontal Angle (θ)
    - Based on the pass-through misunderstanding
  - Vertical Angle (ϕ)
    - Based on the tilt of the mirror

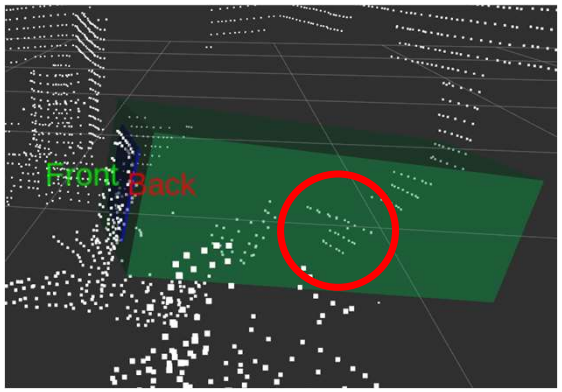
- Shadow Boxing Result
  - Points inside the box are reflected points
    - But some point clouds extend slightly beyond the box
      - ✓ Correct this error by using Kd-Tree



Horizontal Angle



Vertical Angle



Result

Reflected Points Recovery

Mirror Plane Point

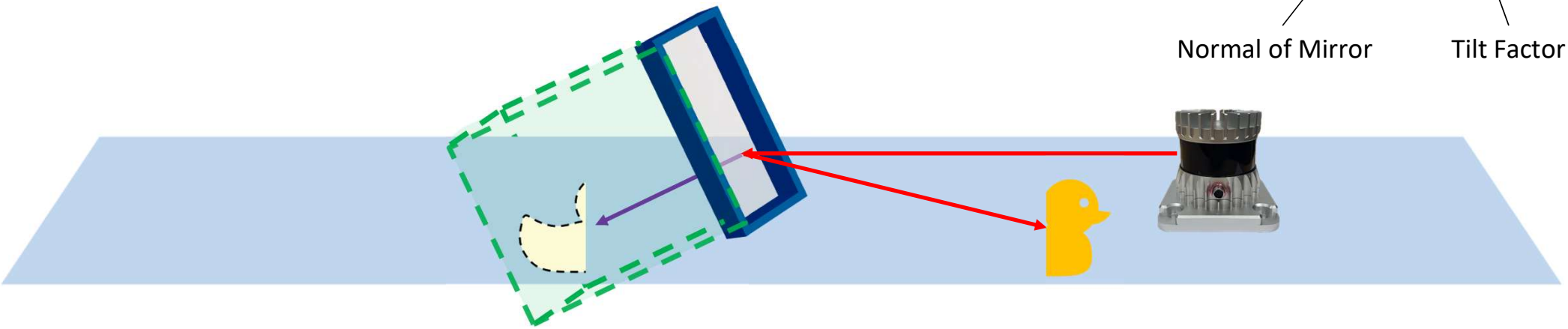
$$\boxed{Restored\ Position:\ P_{restored} = H(P_{orig} - Q) + Q + c}$$

$$\boxed{Original\ Houserholder\ Matrix:\ H = I - 2nn^T}$$

$$\boxed{Tilt\ Correction:\ c = [0\ \ 0\ \ \Delta z]^T = [0\ \ 0\ \ -n_z \cdot k_z]^T}$$

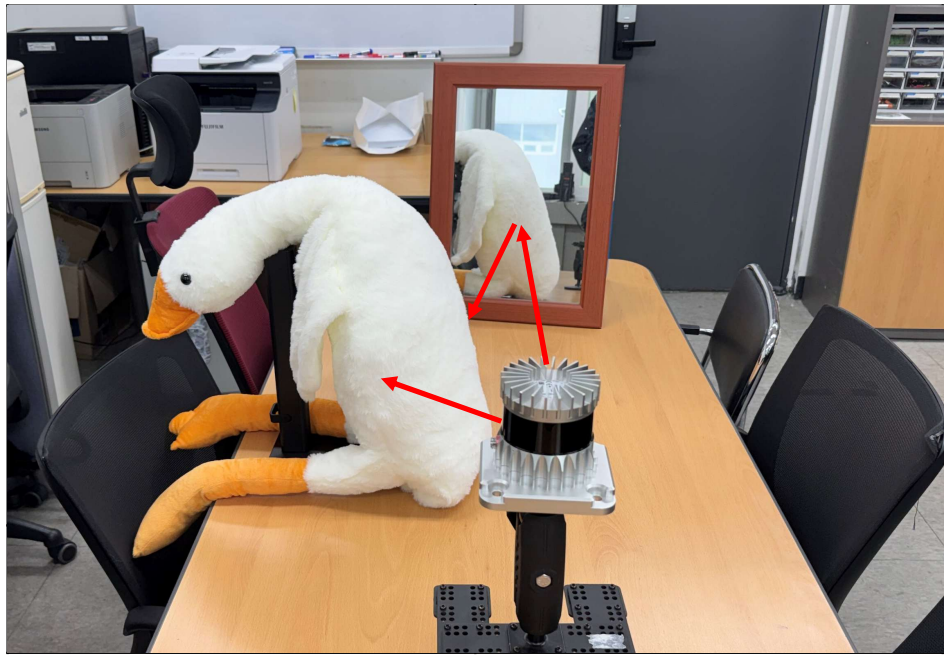
Normal of Mirror

Tilt Factor

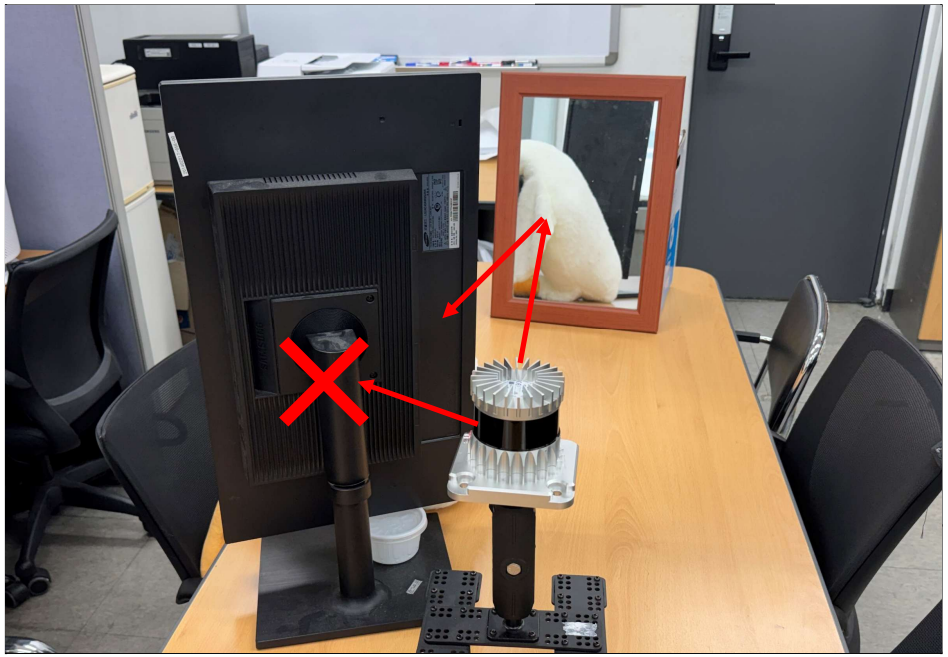


Experiment Setting

- Recovery Experiment
  - Recover complete points of objects

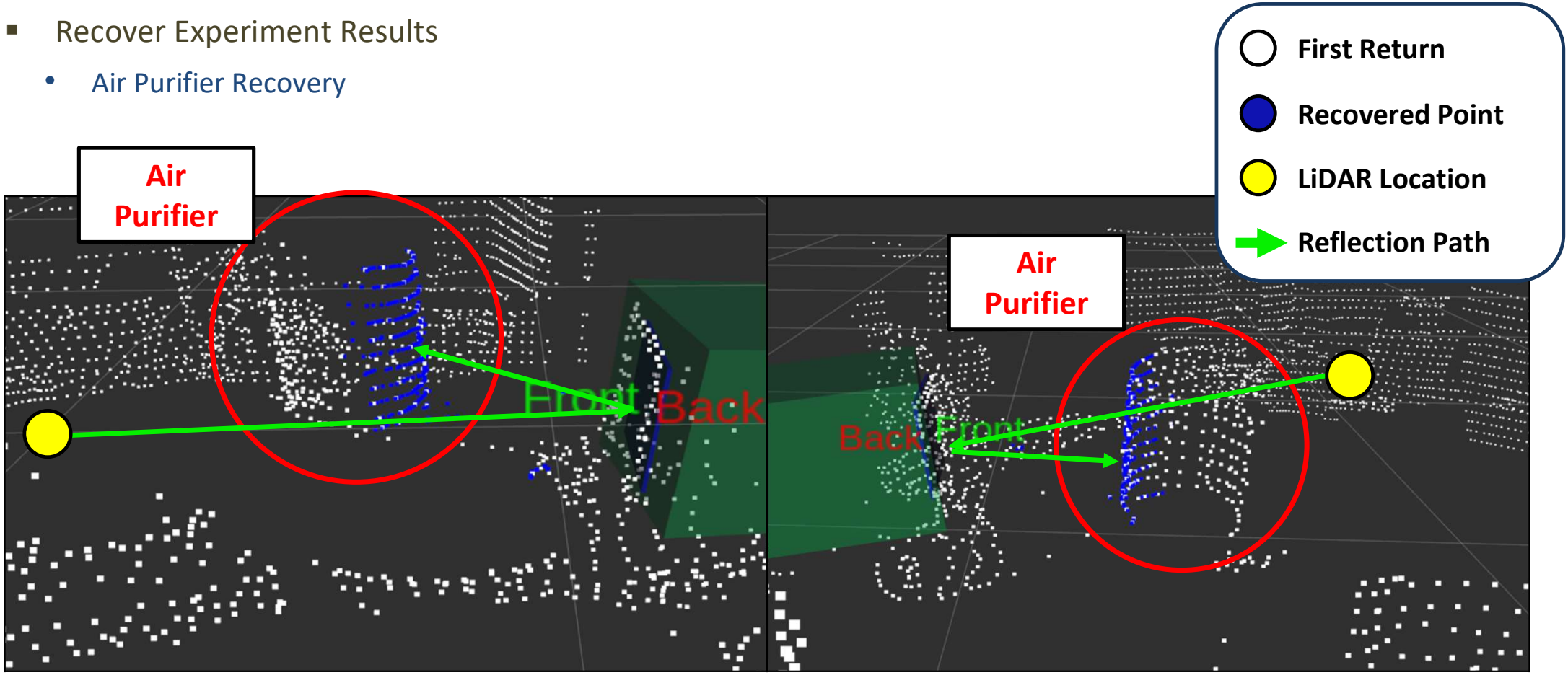


- Obstacle Experiment
  - Detect object behind the obstacle



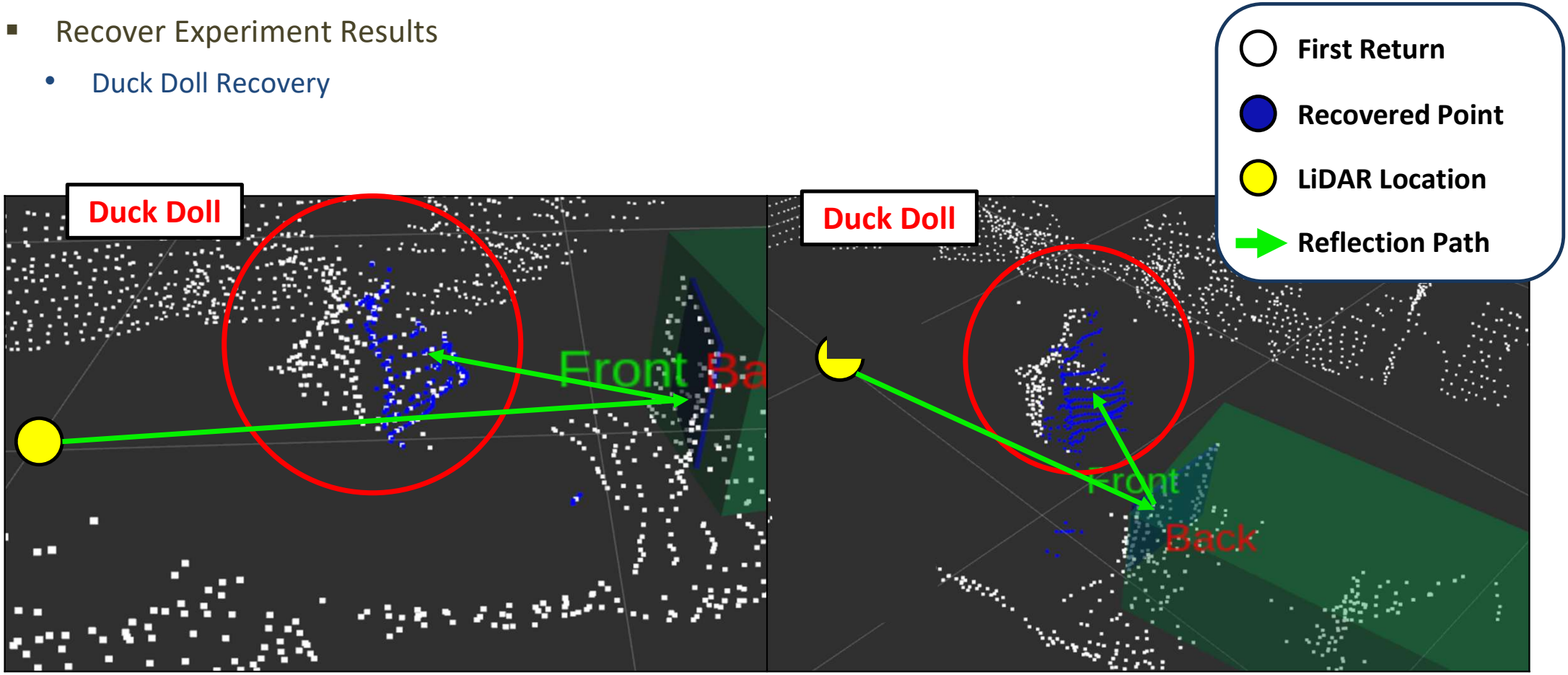
Experiment Results (1)

- Recover Experiment Results
  - Air Purifier Recovery



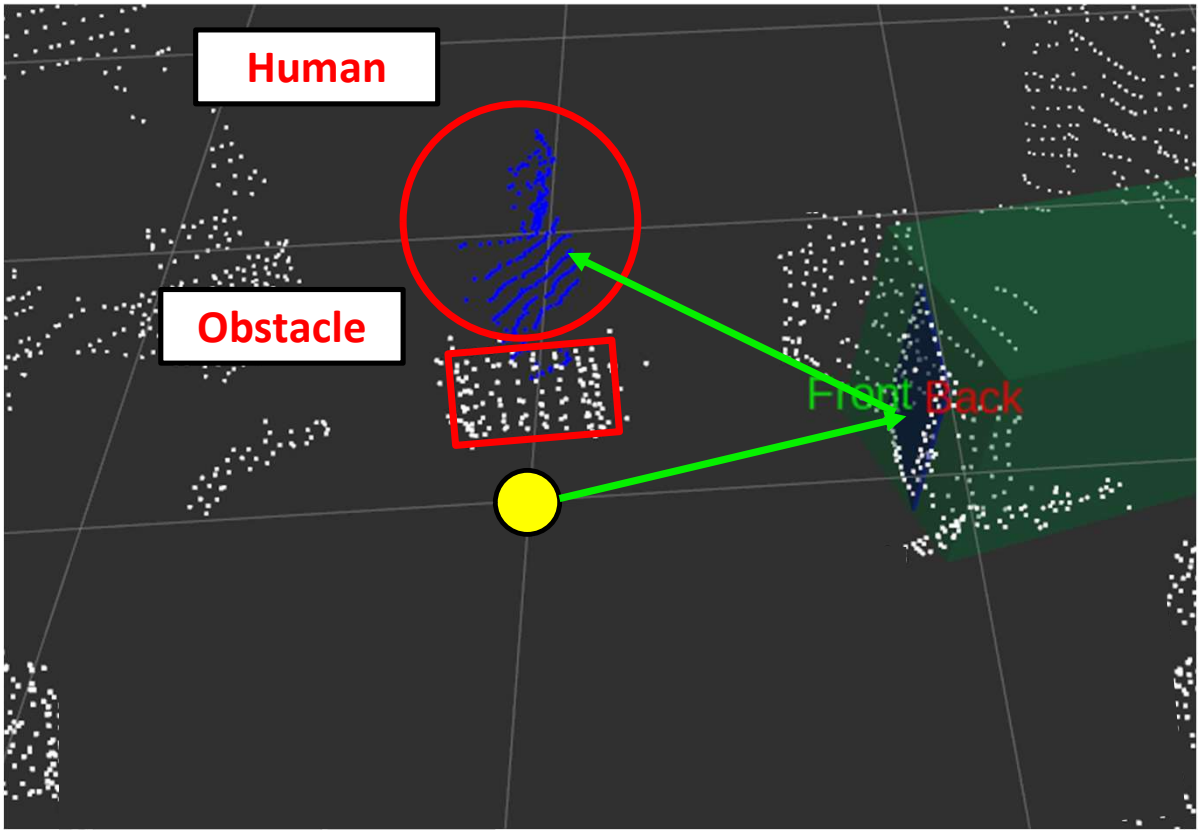
Experiment Results (2)

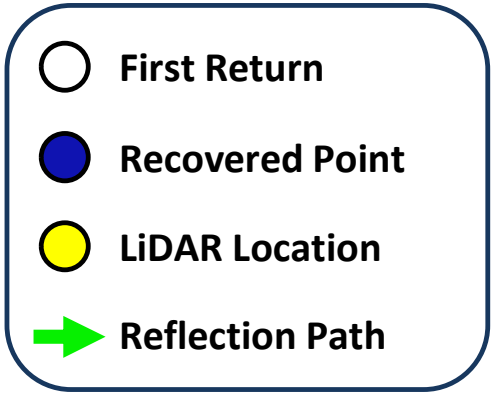
- Recover Experiment Results
  - Duck Doll Recovery



Experiment Results (3)

- Obstacle Experiment Result

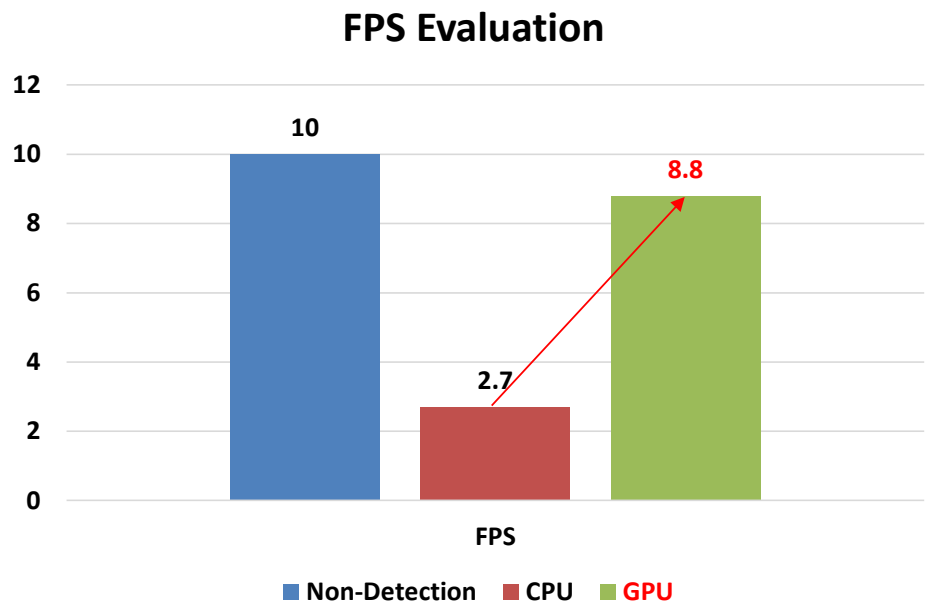




Experiment Results (4)

- Real Time Problem
  - Many vector calculations cause a bottleneck
    - Hard to implement Real-Time Software
  - GPU Acceleration
    - Use Pytorch on the NVIDIA CUDA platform
    - Apply to DBSCAN, RANSAC and Householder Transformation

- FPS Evaluation
  - Hardware Spec
    - Intel Core i7-10700F & NVIDIA GeForce RTX 2060
  - Non-Detection results were the same
  - GPU Acceleration effectively improved FPS



Conclusion

- **Shadow Boxing Method**
  - Mirrors are a challenge for LiDAR sensing
  - Real-Time detection and restoration through Shadow Boxing
  - Z-axis correction is also possible

- **Future works**
  - Multi-modal Sensing to effectively capture the second return
  - Apply our scheme to real vehicular environments

Thank you!

Questions?